Matrix functions and stability

Imaginary-part rank in a limiting Green-function matrix equation

Difficulty: challenging **Importance:** interesting to specialist

Status: conjecture; no general resolution located as of 2026-09-08

Problem statement

Let $n \geq 1$, let $C, D, R, P \in \mathbb{C}^{n \times n}$, and write M^* for conjugate transpose. Suppose $R = R^*$, $P = P^*$, and

$$P + \lambda D^* + \lambda^{-1} D \succ 0 \qquad (|\lambda| = 1).$$

For each $\eta > 0$, let X_η be the unique nonsingular solution of

$$X_\eta + (C^* + i\eta D^*) X_\eta^{-1} (C + i\eta D) = R + i\eta P$$

satisfying $\rho(X_\eta^{-1}(C + i\eta D)) < 1$, where ρ denotes spectral radius. Assume the finite limit $X_0 = \lim_{\eta \downarrow 0} X_\eta$ exists and is nonsingular. Assume the matrix polynomial

$$\mathcal{P}_0(\lambda) = \lambda^2 C^* - \lambda R + C$$

is regular, meaning $\det \mathcal{P}_0(\lambda) \not\equiv 0$. Suppose its eigenvalues on the unit circle are all simple (algebraic multiplicity one). Their number is even; denote it by $2m$.

Is it always true that

$$\operatorname{rank}\left(\frac{X_0 - X_0^*}{2i} \right) = m?$$

Why this matters for NLA

The rank describes the non-Hermitian part of the selected matrix-equation solution in Green-function computations. It connects a limiting nonlinear solve to the spectrum of a structured quadratic polynomial.

The source proves the upper bound by m. Its earlier SIAM paper proves equality for real C, R, $P = I$, $D = 0$; the general complex case remains the question. Existence of the limit is assumed, not conjectured here.

References

1. C.-H. Guo, Y.-C. Kuo, W.-W. Lin, *Numerical solution of nonlinear matrix equations arising from Green's function calculations in nano research*, JCAM 236 (2012), 4166–4180: equation (1), Theorems 3 and 5, and the conjecture on p. 4172. Author manuscript: pp. 1, 5–8.
2. The same authors, *On a nonlinear matrix equation arising in nano research*, SIMAX 33 (2012), 235–262, §3: the real-coefficient case.

Status check

On 2026-09-08, checked both papers' scope and Guo's publication list through 2026. Searches using the JCAM title, authors, "Green rank conjecture," "weakly stabilizing," and 2025/2026 found no general resolution. The source proves only the upper bound in the stated complex setting. This is a bounded literature check.